

TASK 1

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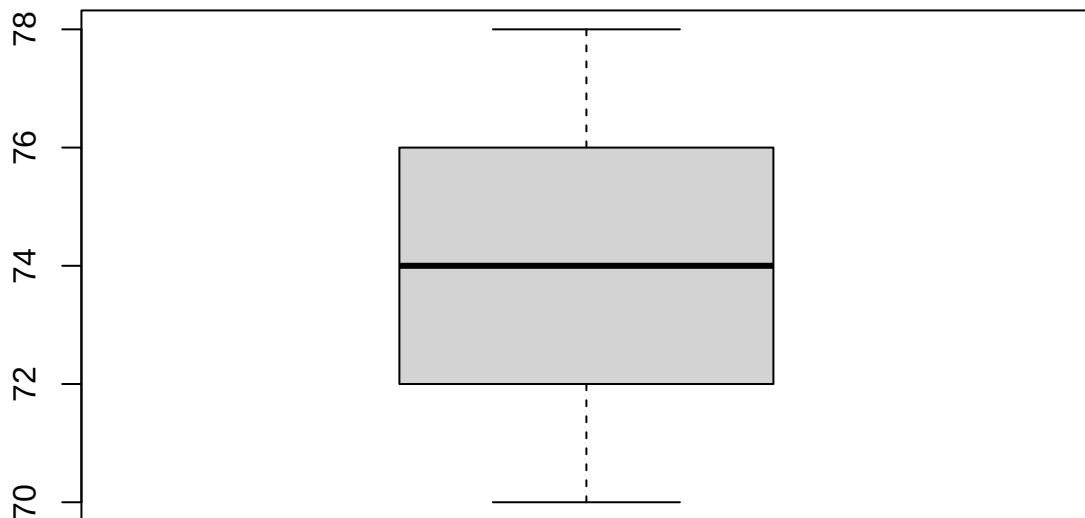
```
traditional=c(72, 75, 78, 70, 74, 77, 73, 76, 71, 74)
new=c(80, 82, 78, 85, 79, 81, 84, 83, 80, 82)
library(psych)
describe(traditional)
```

```
##      vars  n mean   sd median trimmed  mad min max range skew kurtosis  se
## X1      1 10   74 2.58    74      74 2.97  70  78     8    0    -1.41 0.82
```

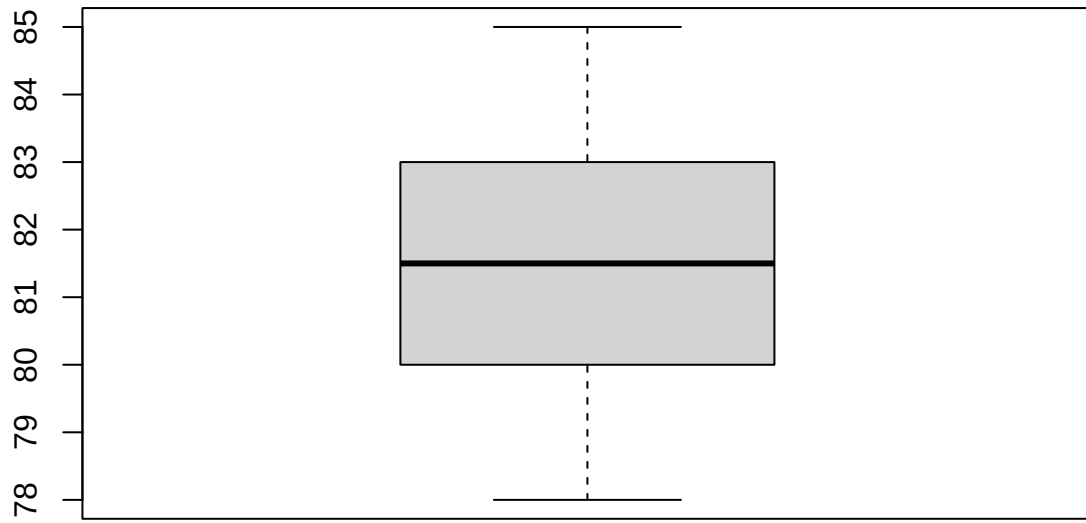
```
describe(new)
```

```
##      vars  n mean   sd median trimmed  mad min max range skew kurtosis  se
## X1      1 10 81.4 2.22   81.5   81.38 2.22  78  85     7 0.09    -1.38 0.7
```

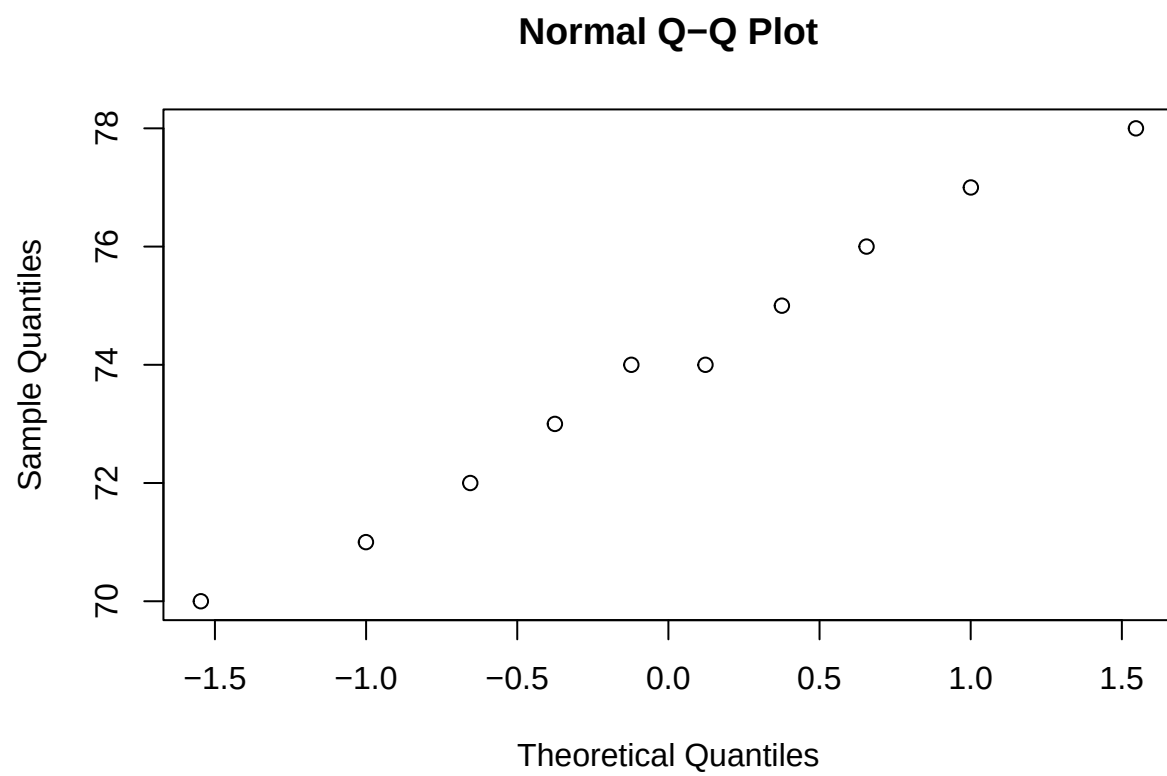
```
boxplot(traditional)
```



```
boxplot(new)
```

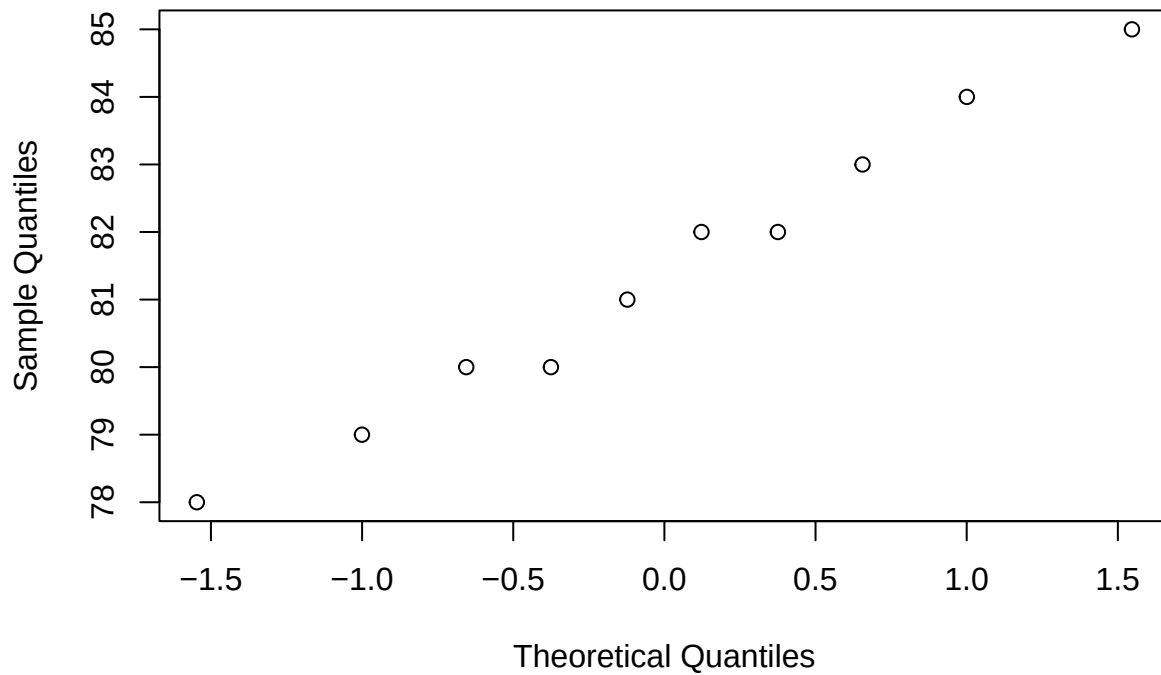


```
qqnorm(traditional)
```



```
qqnorm(new)
```

Normal Q-Q Plot



```
var(traditional)
```

```
## [1] 6.666667
```

```
var(new)
```

```
## [1] 4.933333
```

```
var(traditional)/var(new)
```

```
## [1] 1.351351
```

```
t.test(traditional,new,mu=0,alternative = "less",var.equal = TRUE,conf.level = 0.95)
```

```
##
## Two Sample t-test
##
## data: traditional and new
## t = -6.8707, df = 18, p-value = 9.96e-07
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
##      -Inf -5.532356
## sample estimates:
## mean of x mean of y
##      74.0      81.4
```

the descriptive statistic

the mean is group traditional is 74 and standard deviation equal 2.58 the mean is group new is 81.4 and standard deviation equal 2.22 ## the boxplot is traditional and new is : the mean is traditional group equal 74 and new group is 81.4 As shown, the average for the traditional group equals the average for the new group but, But dispersion is different

the inference statistic

A check for the normality condition was performed, and the results showed that both groups had a normal distribution. Using a drawing q-q plot, The sample size in both groups was small and two groups are independent and the same variance. A test was independent sample t-test used to determine whether the new teaching method improved student performance compared to the traditional method. The results showed a difference between the average scores of the two groups, and that the new teaching method was better than the traditional one. Where the p-value = $9.96e-07 < 0.05$ and $df = 18$ and the sample of traditional is 74 and new is 81.4